HUMIDITY EFFECTS IN SOLIDS DRYING PROCESSES

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Abstract: *Thermal drying from a wet solid to a dry solid is an essential intermediate step in many solids processing plants. Humidity has a major effect on dryer performance and ability to meet quality specifications, particularly for convective dryers. Humidity calculations using a psychrometric chart give an excellent short-cut method for dryer sizing. Ambient and inlet humidity affects both drying kinetics and equilibrium moisture content, especially at lower drying temperatures, and is also important in storage. Exhaust humidity measurement is also useful for tracking and controlling the progress of drying, especially as direct moisture content measurement is often difficult. However, accurate and reliable measurements are difficult due to the hot, damp and dusty environment. The recent revision of British Standard BS 1339 has helped to clarify definitions and improve calculation methods.*

Introduction

Drying of solids

Drying is an essential intermediate step in many manufacturing processes, e.g. for chemicals, foods, paper, textiles, consumer products and miscellaneous particles and powders. It accounts for 10-20% of the total energy consumption of many countries, because of the heat that must be supplied corresponding to the latent heat of evaporation. The vapour produced emerges as a high humidity exhaust stream. Usually the liquid being removed is water, but it is also common to be evaporating an organic solvent.

Most drying processes are either convective (with heat supplied by hot air) or conductive (with heat transmitted from hot walls, pipes or surfaces). Radiation (e.g. infra-red), dielectric heating (radiofrequency and microwaves) or combinations may also be used.

Humidity and its definition

Humidity is a key factor in drying processes. It expresses the amount of vapour in the carrier gas stream (usually air). However, there is often confusion between relative and absolute humidity. Absolute humidity is a measure of the mass of moisture in the air (g/kg, kg/kg or g/m^3); relative humidity (RH) is a percentage of the saturation humidity at the given temperature. Either may be important in different situations. For example, absolute humidity dictates the heat and mass balance, while relative humidity influences the equilibrium moisture content. However, a change in temperature has a major effect on RH, whereas absolute humidity is unchanged if it is quoted in mass units (e.g. g/kg). Ideally, both instrument displays and plant control rooms should show both relative and absolute humidity.

Even for absolute humidity, there is confusion over terminology. Often, it is defined as a volumetric concentration in g/m^3 or kg/m^3. This is acceptable in fields such as meteorology and air-conditioning which deal with air-water systems close to atmospheric pressure and ambient temperature. However, it is very inconvenient for dryer calculations because the volume varies too much over the wide range of temperatures encountered. Hence, in drying and other chemical engineering processes, "absolute humidity" is usually defined on a mass basis, and expressed in units of kg (moisture) per kg (dry gas). In other disciplines this is often called the "mixing ratio". Users should always look at the stated units when interpreting humidity data!

Likewise, confusion often arises because volumetric flowrates differ substantially depending on whether they are stated at operating temperature (maybe 100-200°C, or even higher) or ambient temperature. Even for so-called "standard" flows (e.g. SCFM, standard cubic feet per minute) several different values are used for the "standard" temperature (ranging from 0°C to 25°C), and calculation errors can result. Hence, in drying calculations it is generally best to use mass flowrate in kg/s (or kg/h) and mass velocity in kg/m^{2}s (mass flux) in preference to volumetric flowrate (m^3/s) and velocity (m/s) respectively. Air flowmeters should preferably display both volumetric and mass flowrates

Dewpoint is directly related to the absolute humidity (and total pressure); for example, an air-water mixture with 7.5 g water vapour per kg of air has a dewpoint of approximately 10°C (at typical atmospheric pressures).

An important step forward in reducing confusion and standardising best practice is the recent updating of British Standard BS 1339 on Humidity. Part 1 (2002) contains rigorous definitions and formulae, both for the air-water system and other solvent-gas systems, including interconversions between all the parameters given above. Part 2 (due for publication 2007) covers calculation methods, based on a spreadsheet with inbuilt functions for all the key humidity transformations. This allows users to generate calculations or tables for any desired set of conditions, and replaces several reams of lookup tables. Finally, Part 3 (2004) is a practical descriptive guide to the measurement of humidity and dewpoint.

Humidity effects in dryers

Humidity affects the dryness of the solids it is in contact with. A dynamic equilibrium is set up between the moisture in the solids and the vapour in the air. Solids left in contact with moist air for long periods will reach their equilibrium moisture content, X_E, which increases as relative humidity increases or temperature falls. As a result, solids in a cool damp atmosphere will pick up more moisture than those stored in warm dry conditions. X_E is often plotted as an adsorption or desorption isotherm.

Equilibrium relative humidity (ERH) uses the principle in reverse.

For a given solids moisture content, the relative humidity of the air in contact with the damp solid at equilibrium will be constant. This provides a useful indirect method of measuring solids moisture content.

Temperature or humidity cycling often affects solids in storage. For example, the air in an uncontrolled storage area will be at a relatively high temperature and absolute humidity during the day. At night the temperature falls and the relative humidity increases until the dewpoint is reached and air condenses on the walls and the solid surface ("silo rain"). This surface moisture then cakes the solid together into large lumps so that it will not flow freely on discharge. A similar problem can arise if solids are discharged from the dryer without cooling, and stored or bagged when hot. Further moisture is evaporated, giving high local humidity; when the air cools down to ambient, condensation again occurs. Humidity should always be carefully considered whenever there are caking and lump formation problems in storage and transport. High relative humidity may also encourage product spoilage, e.g. by promoting biological reactions and mould growth.

Applying Humidity In Calculations

The needs of the practitioner involved in industrial drying are completely different to those of the standards expert. He is not generally interested in predicting or measuring values of humidity and temperature accurate to several decimal places; in the challenging environment of a dryer exhaust he is pleased to have any instrument which keeps working and gives a reading accurate to the nearest degree or a few % RH. Humidity helps with shortcut calculations for dryer design and performance and in identifying trends as operating conditions vary.

Psychrometric charts and dryer sizing

The familiar Grosvenor temperature-humidity psychrometric chart is not the only one used in drying. Enthalpy-humidity psychrometric charts are also extremely useful. The enthalpy plotted is the total of the sensible heat and latent heat of the gas-vapour mixture. There are two types; Mollier (non-orthogonal, used in UK and Europe) and Bowen (orthogonal, used in USA). The constant-enthalpy lines on the Mollier chart slope downwards with a gradient proportional to the latent heat. Though less intuitive initially, it is clearer to read in practice as the information tends to be cramped together on the Bowen chart. The charts provide a useful visual representation of gas-phase conditions during drying processes.

Psychrometric charts are very useful for rapid sizing of convective (air heated) dryers, by finding what airflow will be needed. The sensible heat lost from the gas as it is cooled becomes latent heat of vapour which returns to the mixture, and the **total** gas-vapour mixture enthalpy remains roughly constant. Hence, the air conditions during drying roughly follow a constant-enthalpy line. This provides a simple visual way of determining dryer outlet conditions and performing a short-cut design calculation. Allowing a suitable safety margin above the dewpoint, say 20^{0}C, the exhaust temperature and the corresponding humidity can be read off from the chart. A simple mass balance on the moisture in the gas gives the required airflow.

For example, suppose we want to dry 1 tonne/hr of wet solid from 15% moisture to 3%. The required evaporation rate is 12% of 1 te/h, or 120 kg/h. If we use air at 150^{0}C with an inlet humidity of 10 g/kg, the chart (Figure 1) shows the wet-bulb temperature is about 42^{0}C. Assuming an outlet temperature of 60^{0}C, we find that the dewpoint is 40^{0}C and exhaust humidity 46 g/kg. The humidity change over the dryer is 36 g/kg (0.036 kg/kg) so to evaporate 120 kg/h we will need (120/0.036) or 3330 kg/h of air. A safety margin of about 10% can be added to allow for heat losses.

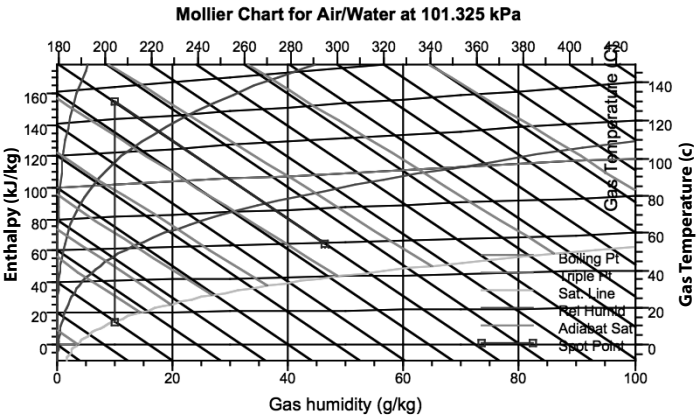

Figure 1 Mollier psychrometric chart with typical operating line for adiabatic dryer (Courtesy Aspen Technology Inc.)

Temperature driving force and drying rate

If surface moisture is present, the outer surface of a solid will be at the wet-bulb temperature T_{wb} as long as any surface moisture is present. T_{wb} is affected by humidity as well as temperature, and the driving force for drying is the difference between dry-bulb and wet-bulb temperatures, (T_g-T_{wb}).

Obtaining wet-bulb temperature from dry-bulb temperature and humidity previously required a tedious iterative calculation. The availability of functions such as those in BS 1339 Part 2 is a major help, and it is also convenient to have direct approximate relationships for wet-bulb temperature based on inlet temperature and humidity. The best fit depends on the temperature and humidity range selected. For example, for pharmaceutical drying, the range of interest is typically 40-80^{0}C inlet temperature and 0.005-0.015 g/kg absolute humidity (Y). The following relationship gives T_{wb} to within 1^{0}C over this range and, equally important, gives the driving force (T_g-T_{wb}) to within 1%:

$$T_{wb} = 0.24T_g + 500Y + 8$$

However, at the higher inlet temperature ranges typically found for dryers in the chemical industry, the following fit applies for $80<T_g<150$ and $Y<0.02$ kg/kg:

$$T_{wb} = 3.18\sqrt{(T_g + 2400Y_i)}$$

How much difference does humidity make to wet-bulb temperature and driving force? The answer is that it depends on temperature; it is a small effect for a dry-bulb temperature of 150^{0}C, but is significant at 80^{0}C and substantial at 40^{0}C.

Table 1 has been generated using the humidity spreadsheet functions in draft British Standard BS 1339 Part 2, and shows that if the dewpoint changes from 10^{0}C to 20^{0}C at an inlet temperature of 150^{0}C it makes only a 2% difference in driving force (and hence drying time). However, at 80^{0}C the difference is 7% and at 40^{0}C, no less than 26%. Moreover, if one is drying close to equilibrium moisture content and the product ERH is significant – say 50% - an increase in

inlet humidity will increase exhaust humidity and RH substantially, and make drying even slower.

Absolute humidity	g/kg	2	7.5	15	20
Absolute humidity	kg/kg	0.002	0.0075	0.015	0.02
Dewpoint T_{dew}	^{0}C	-7	10	20	25
Dry-bulb temperature:		150	^{0}C		
Wet-bulb temperature		41.1	42.8	45.1	46.4
Driving force (T_g-T_{wb})		108.9	107.2	104.9	103.6
Difference from T_{dew}=10		1.7%	0.0%	-2.1%	-3.3%
Dry-bulb temperature:		80	^{0}C		
Wet-bulb temperature		28.3	31.1	34.6	36.6
Driving force (T_g-T_{wb})		51.7	48.9	45.4	43.4
Difference from T_{dew}=10		5.9%	0.0%	-7.0%	-11.1%
Dry-bulb temperature:		40	^{0}C		
Wet-bulb temperature		16.6	21.0	25.9	28.6
Driving force (T_g-T_{wb})		23.4	19.0	14.1	11.4
Difference from T_{dew}=10		23.0%	0.0%	-25.7%	-40.2%

Table 1 Wet-bulb temperature and driving force for varying humidity and temperature

Inlet And Outlet Humidity In Dryers

In dryers, inlet humidity is a controlled parameter which affects the process, whereas outlet humidity is a monitoring parameter which is affected by the process.

Inlet humidity

Variations in inlet humidity will naturally occur as the ambient conditions fluctuate, with day/night and winter/summer variations. For high drying temperatures (e.g. 150^{0}C or above) the effect on drying is generally small. However, at the lower temperatures (less than 100^{0}C) used for temperature-sensitive materials, such as foods and pharmaceuticals, changes in inlet humidity significantly affect the wet-bulb temperature (and hence driving forces and drying times), and the exhaust humidity (influencing product moisture content). Therefore, inlet humidity control must be considered, with the following options:

- No control. If the effect on the drying process is unimportant.
- Capping. A chilled water coil is used to ensure the inlet dewpoint is no higher than a certain value, e.g. 10^{0}C (7-8 g/kg).
- Dehumidification. If very dry air is needed for moisture-sensitive materials, a chemical absorption wheel can be used to give humidities of 2 g/kg or less.
- Rehumidification. If the product could dry out too fast or too far, steam or water injection can be used to increase humidity.

High humidities can also be achieved by recycling exhaust air. For example, when drying leather in tunnel dryers, air recycle is used to maintain relative humidity in the circuit at 40-50% at temperatures of 40-80°C. Leather cracks and denatures if dried too far, and these conditions keep the equilibrium moisture content at about 15% and prevent further drying.

To control humidity within a tight range (e.g. 5 to 8 g/kg), a combination of capping and rehumidification can be used.

Exhaust humidity

Outlet humidity is a good indicator of the progress of drying. For continuous drying, exhaust humidity monitoring shows whether drying is consistent, and can give early warning of fluctuations in solids inlet moisture content or mass flowrate (which can be difficult to measure accurately on-line). For batch drying, exhaust humidity can give an indirect measure of how close the solids are to their desired final moisture content (endpoint).

Outlet moisture content is a key part of the dryer product specification, and ideally should be measured on-line. For continuous dryers, this can be done using near infra-red on the product stream, but batch dryers present a greater problem. In-vessel measurements suffer from fouling and non-uniform solids mixing. Hence product moisture is normally deduced from indirect measurements. Product or exhaust gas temperature are normally used as indicators of the endpoint, but exhaust humidity provides a useful alternative or additional information.

Figure 2 shows results from a drying kinetics rig at SPS, Harwell, where the outlet humidity was monitored using a fast-response spectroscopic infra-red gas analyser. A mass balance over the dryer module is used to convert humidity to evaporation rate from the solids, giving Figure 3. Then, integration over time gives the drying curve (moisture content variation with time), Figure 4. Note that the integration smoothes out the scatter on the humidity data. However, the result is sensitive to errors in flowrate and humidity measurement, so it is essential to cross-check the drying curve with the initial and final solids moisture contents obtained from an accurate laboratory method, usually an oven test. The mass balance should agree to within 5%.

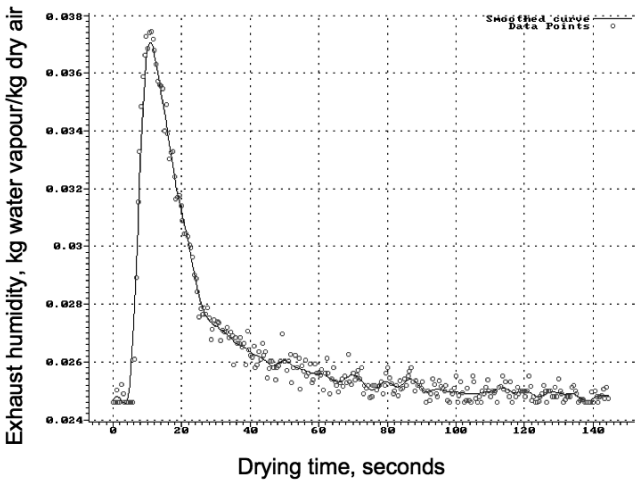

Figure 2 Experimental humidity-time plot

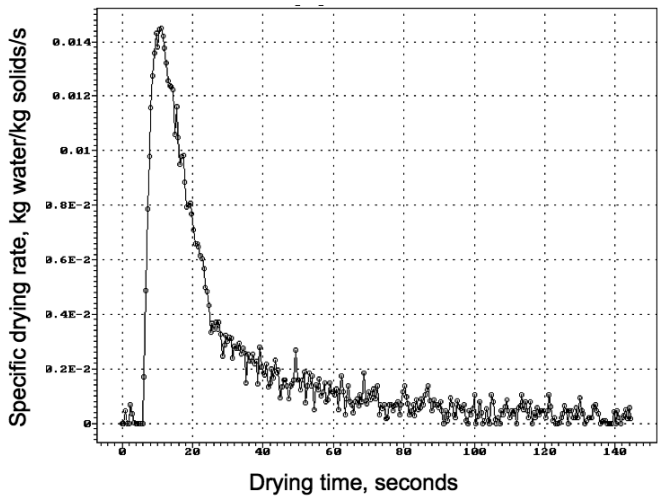

Figure 3 Derived plot of drying rate against time

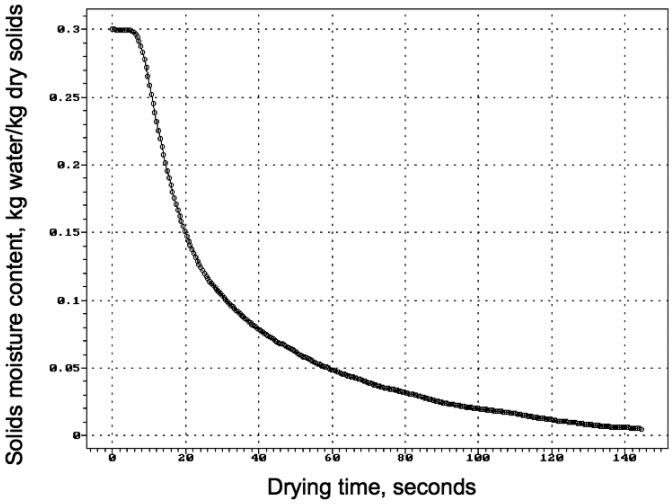

Figure 4 Drying curve (moisture content v. time)

Measurement And Control

Dryer control

A modern well-instrumented dryer should have sensors for air temperature, humidity and flowrate, and solids temperature and (where possible) moisture content, as shown in Figure 5. Also, for continuous dryers, solids flowrate should be monitored.

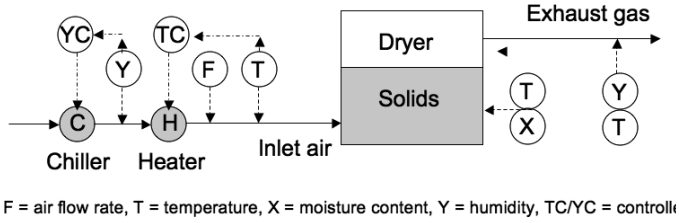

Figure 5 Instrumentation for typical dryer

Direct feedback loops are normally used to control inlet air temperature and, less commonly, inlet air humidity. Exhaust air temperature and humidity, and product temperature and solids moisture content, show how drying is progressing and give warning of any fluctuation. If they vary from their setpoint, inlet temperature can be altered. Automated feedback is possible, but as drying is a complex process and reliable modelling is difficult, manual operator input is more common. For dryers with a long solids residence time, an excursion in outlet humidity and temperature can provide early warning of problems and allow inlet conditions to be varied to restore the solids to the desired product conditions before they emerge.

Humidity measurement

Inlet humidity is relatively easy to measure. The airflow is clean (often filtered) and at relatively low humidity. A sensor placed after the heater may have to work at high temperature and low relative humidity (1% or less), or one after a chiller (to maintain a given dewpoint) may be at 100% saturation, but it is not difficult to procure reliable and inexpensive instruments for these conditions. For example, capacitance RH sensors or chilled-mirror dewpoint sensors can be used.

In contrast, measurement of outlet (exhaust) humidity has been a difficult challenge for many years. The hot, wet, dusty environment has frequently led to instrument fouling and breakdown. The ideal specification for an exhaust humidity sensor would be:

- Robust; able to deal with near-saturation conditions at moderate temperatures, or full dryer inlet temperature
- Reliable; able to operate for several months with little or no maintenance
- Reproducible; giving consistent output without needing regular recalibration
- Resistant to or protected from dust fouling
- Rapid response, preferably in seconds rather than minutes, to allow effective monitoring and endpoint detection in batch dryers.

The requirements may be conflicting, for example, a perforated dust shield around a sensor can slow down response time.

Exhaust humidity sensors should preferably be mounted after air filters which collect the worst of the dust, but some fine dust penetration is inevitable.

Effective use of exhaust humidity for control and monitoring has been hamstrung by a lack of proven, robust, reliable sensors. Comparing this with the complexity of control software, Stuart Gardiner (ex ICI) commented in the 1990's; "The brain of the plant has outstripped its eyes and ears". Although modern RH capacitance and chilled mirror sensors are considerably improved over previous versions, there is still a greater need for verification and testing in practical situations, particularly the exhaust of real industrial dryers.

Output should be in the form of absolute humidity or dewpoint; therefore, for a sensor measuring relative humidity, a co-located temperature measurement is also needed.

Other Aspects

Humidity does not simply apply to water vapour in air. Many drying processes involve the removal of solvents, and the same considerations apply for drying kinetics and equilibria. BS1339 Part 1 provides generalised equations for any solvent-gas system.

In some cases, trace quantities of a solvent are being removed as well as large quantities of water, and even a small solvent concentration in the drying gas may prevent the reduction of the solvent component to the very low levels frequently required in product specifications. In a gas recycle system, even with a condenser on the loop, a trace organic component may build up as its concentration is too low to be condensed out at the condenser temperature appropriate to recovery of the main component (water). In one case, the purge rate had to be increased from an original estimate of 5-10% to at least 20% to keep a solvent below a concentration of 0.25% in the final product.

Humidity requirements can also affect dryer selection. The curves in Figure 2-4 were for a crossflow dryer where the inlet gas contacting the solids is at the same temperature throughout the dryer. However, in co-current dryers, the gas temperature falls steadily throughout the dryer. This is good for heat-sensitive materials (dry solids are only exposed to cool gas and are unlikely to overheat) but the cool wet exhaust gives a high equilibrium moisture content and this may make low outlet moisture contents difficult to obtain. Conversely, if a very low outlet moisture content is desired, a countercurrent dryer may be used; the exiting solids will be in contact with the hot dry inlet gas, minimising product moisture and maximising local driving forces for drying. The problem here can come at the inlet; the wet gas contacts the cold incoming solids and can condense on the surface, which may give local stickiness problems.

Conclusions

Humidity is a key concept in understanding and controlling industrial drying processes. Despite recent advances in instrumentation, there is a continuing need for proven, robust and reliable sensors, particularly for exhaust humidity measurement.